

Experiment/Case Study No : 6

Aim:

Use string functions and regex for email validation, data extraction, and text analysis.

Software/Tools Required

- Visual Studio Code
- Google Chrome (Web Browser)

Experiment Program Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Detector</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="container">
    <h2>Email Validator & Text Reverser</h2>

    <div class="tool-section">
      <h3>Email Validation</h3>
      <div class="input-group">
        <label for="emailInput">Enter Emails (comma or space separated):</label>
        <input type="text" id="emailInput" placeholder="user@example.com,
test@domain.org">
      </div>
      <button onclick="processEmails()" class="btn-email">Validate Emails</button>
      <div id="emailResults" class="result-box"></div>
    </div>

    <hr class="divider">

    <div class="tool-section">
      <h3>Reverse Paragraph</h3>
      <div class="input-group">
        <label for="reverseInput">Enter Text to Reverse:</label>
        <input type="text" id="reverseInput" placeholder="Type your paragraph
here...">
      </div>
      <button onclick="reverseText()" class="btn-reverse">Reverse Text</button>
      <div id="reverseResults" class="result-box"></div>
    </div>

    <hr class="divider">

    <div class="tool-section">
      <h3>Vowel Counter</h3>
      <div class="input-group">
        <label for="vowelInput">Enter Paragraph to Count Vowels:</label>
        <input type="text" id="vowelInput" placeholder="Type your paragraph
here...">
      </div>
      <button onclick="countVowels()" class="btn-email" style="background: linear-
gradient(45deg, #11998e, #38ef7d);">Count Vowels</button>
      <div id="vowelResults" class="result-box" style="border-left-color: #38ef7d;
```

```

background: linear-gradient(90deg, rgba(56, 239, 125, 0.1), transparent);"></div>
</div>
</div>
<script>
const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;

function processEmails() {
  const input = document.getElementById('emailInput').value;
  const resultsDiv = document.getElementById('emailResults');

  resultsDiv.innerHTML = "";
  resultsDiv.style.display = 'none';

  if (!input.trim()) {
    resultsDiv.innerHTML = '<p class="error">Please enter email addresses.</p>';
    resultsDiv.style.display = 'block';
    return;
  }

  const validEmails = [];
  const invalidEmails = [];
  const potentialEmails = input.split(/[s,]+/);

  potentialEmails.forEach(email => {
    if (email.trim() === "") return;
    if (emailRegex.test(email)) {
      validEmails.push(email);
    } else {
      invalidEmails.push(email);
    }
  });

  const domainCounts = {};
  validEmails.forEach(email => {
    const domain = email.split('@')[1];
    domainCounts[domain] = (domainCounts[domain] || 0) + 1;
  });

  let output = '<h4>Analysis Results</h4>';
  output += '<div class="stat-grid">';
  output += '<div class="stat-item"><strong>Valid:</strong> <span class="count-green">' + validEmails.length + '</span></div>';
  output += '<div class="stat-item"><strong>Invalid:</strong> <span class="count-red">' + invalidEmails.length + '</span></div>';
  output += '</div>';

  if (validEmails.length > 0) {
    output += '<p><strong>Valid Emails:</strong></p><ul class="email-list valid">';
    validEmails.forEach(e => output += '<li>' + e + '</li>');
    output += '</ul>';
  }

  if (invalidEmails.length > 0) {
    output += '<p><strong>Invalid Entries:</strong></p><ul class="email-list invalid">';
    invalidEmails.forEach(e => output += '<li>' + e + '</li>');
    output += '</ul>';
  }

  if (Object.keys(domainCounts).length > 0) {
    output += '<p><strong>Domains:</strong></p><ul class="domain-list">';
    for (const domain in domainCounts) {
      output += '<li>' + domain + ' (' + domainCounts[domain] + ')</li>';
    }
    output += '</ul>';
  }
}

```

```

    }

    resultsDiv.innerHTML = output;
    resultsDiv.style.display = 'block';
}

function reverseText() {
    const input = document.getElementById('reverseInput').value;
    const resultsDiv = document.getElementById('reverseResults');

    resultsDiv.innerHTML = "";
    resultsDiv.style.display = 'none';

    if (!input.trim()) {
        resultsDiv.innerHTML = '<p class="error">Please enter text to reverse.</p>';
        resultsDiv.style.display = 'block';
        return;
    }

    const reversedString = input.split("").reverse().join("");

    let output = '<h4>Reversed Output</h4>';
    output += '<div class="reverse-item"><strong>Full Reverse:</strong><br><span' +
class="code-text">' + reversedString + '</span></div>';

    resultsDiv.innerHTML = output;
    resultsDiv.style.display = 'block';
}

function countVowels() {
    const input = document.getElementById('vowelInput').value;
    const resultsDiv = document.getElementById('vowelResults');

    resultsDiv.innerHTML = "";
    resultsDiv.style.display = 'none';

    if (!input.trim()) {
        resultsDiv.innerHTML = '<p class="error">Please enter text to count
vowels.</p>';
        resultsDiv.style.display = 'block';
        return;
    }

    const vowels = input.match(/[aeiou]/gi);
    const vowelCount = vowels ? vowels.length : 0;

    const vowelCounts = { a: 0, e: 0, i: 0, o: 0, u: 0 };
    input.toLowerCase().split("").forEach(char => {
        if (vowelCounts.hasOwnProperty(char)) {
            vowelCounts[char]++;
        }
    });

    let output = '<h4>Vowel Analysis</h4>';
    output += '<div class="stat-grid">';
    output += '<div class="stat-item"><strong>Total Vowels:</strong> <span' +
class="count-green">' + vowelCount + '</span></div>';
    output += '</div>';

    output += '<p><strong>Breakdown:</strong></p><ul class="domain-list">';
    for (const vowel in vowelCounts) {
        output += '<li>' + vowel.toUpperCase() + ': ' + vowelCounts[vowel] + '</li>';
    }
    output += '</ul>';

    resultsDiv.innerHTML = output;

```

```

        resultsDiv.style.display = 'block';
    }
</script>
</body>
</html>

```

```

body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    background: linear-gradient(135deg, #0f0c29, #302b63, #24243e);
    margin: 0;
    padding: 40px 20px;
    color: #e0e0e0;
    min-height: 100vh;
}

.container {
    max-width: 800px;
    margin: 0 auto;
    background: rgba(255, 255, 255, 0.05);
    backdrop-filter: blur(10px);
    padding: 40px;
    border-radius: 20px;
    box-shadow: 0 8px 32px 0 rgba(0, 0, 0, 0.37);
    border: 1px solid rgba(255, 255, 255, 0.1);
}

h2 {
    text-align: center;
    color: #ffffff;
    text-shadow: 0 0 10px rgba(255, 255, 255, 0.5);
    margin-top: 0;
    letter-spacing: 2px;
}

h3 {
    color: #a8c0ff;
    margin-bottom: 15px;
    font-size: 1.4em;
    text-shadow: 0 0 5px rgba(168, 192, 255, 0.3);
}

.tool-section {
    margin-bottom: 20px;
}

.input-group {
    margin-bottom: 20px;
}

label {
    font-weight: 600;
    display: block;
    margin-bottom: 10px;
    color: #bdc3c7;
    letter-spacing: 1px;
}

input[type="text"] {
    width: 100%;
    padding: 15px;
    border: 1px solid rgba(255, 255, 255, 0.2);
    border-radius: 8px;
    font-size: 16px;
    box-sizing: border-box;
    transition: all 0.3s;
}

```

```

        background: rgba(0, 0, 0, 0.3);
        color: #fff;
    }

    input[type="text"]:focus {
        border-color: #00d2ff;
        outline: none;
        box-shadow: 0 0 15px rgba(0, 210, 255, 0.3);
        background: rgba(0, 0, 0, 0.5);
    }

    button {
        width: 100%;
        padding: 15px;
        border: none;
        border-radius: 8px;
        font-size: 16px;
        font-weight: bold;
        cursor: pointer;
        transition: all 0.3s, transform 0.1s;
        color: white;
        text-transform: uppercase;
        letter-spacing: 1.5px;
        position: relative;
        overflow: hidden;
    }

    button:active {
        transform: scale(0.98);
    }

    .btn-email {
        background: linear-gradient(45deg, #00c6ff, #0072ff);
        box-shadow: 0 4px 15px rgba(0, 114, 255, 0.4);
    }

    .btn-email:hover {
        background: linear-gradient(45deg, #0072ff, #00c6ff);
        box-shadow: 0 6px 20px rgba(0, 114, 255, 0.6);
        transform: translateY(-2px);
    }

    .btn-reverse {
        background: linear-gradient(45deg, #f12711, #f5af19);
        box-shadow: 0 4px 15px rgba(241, 39, 17, 0.4);
    }

    .btn-reverse:hover {
        background: linear-gradient(45deg, #f5af19, #f12711);
        box-shadow: 0 6px 20px rgba(241, 39, 17, 0.6);
        transform: translateY(-2px);
    }

    .divider {
        border: 0;
        height: 1px;
        background: linear-gradient(90deg, transparent, rgba(255, 255, 255, 0.3),
transparent);
        margin: 50px 0;
    }

    .result-box {
        margin-top: 30px;
        padding: 25px;
        border-radius: 12px;
        background: rgba(0, 0, 0, 0.2);
    }

```

```

    border-left: 5px solid #ccc;
    display: none;
    animation: fadeIn 0.5s ease;
    backdrop-filter: blur(5px);
}

#emailResults {
    border-left-color: #00d2ff;
    background: linear-gradient(90deg, rgba(0, 210, 255, 0.1), transparent);
    box-shadow: 0 0 15px rgba(0, 210, 255, 0.1);
}

#reverseResults {
    border-left-color: #f5af19;
    background: linear-gradient(90deg, rgba(245, 175, 25, 0.1), transparent);
    box-shadow: 0 0 15px rgba(245, 175, 25, 0.1);
}

@keyframes fadeIn {
    from { opacity: 0; transform: translateY(10px); }
    to { opacity: 1; transform: translateY(0); }
}

h4 {
    margin-top: 0;
    color: #fff;
    border-bottom: 1px solid rgba(255, 255, 255, 0.1);
    padding-bottom: 15px;
    text-transform: uppercase;
    font-size: 0.9em;
    letter-spacing: 1px;
}

.stat-grid {
    display: flex;
    justify-content: space-around;
    margin-bottom: 20px;
    background: rgba(255, 255, 255, 0.05);
    padding: 15px;
    border-radius: 8px;
    box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);
    border: 1px solid rgba(255, 255, 255, 0.05);
}

.stat-item {
    text-align: center;
    font-size: 1.1em;
    color: #bdc3c7;
}

.count-green {
    color: #00ff88;
    font-weight: bold;
    text-shadow: 0 0 10px rgba(0, 255, 136, 0.4);
}

.count-red {
    color: #ff4b4b;
    font-weight: bold;
    text-shadow: 0 0 10px rgba(255, 75, 75, 0.4);
}

ul {
    padding-left: 0;
    margin: 15px 0;
}

```

```
.email-list.valid li {
  color: #00ff88;
  font-family: 'Courier New', monospace;
  background: rgba(0, 255, 136, 0.05);
  padding: 10px;
  margin-bottom: 8px;
  border-radius: 6px;
  list-style-type: none;
  border: 1px solid rgba(0, 255, 136, 0.2);
  transition: all 0.2s;
}

.email-list.valid li:hover {
  background: rgba(0, 255, 136, 0.1);
  transform: translateX(5px);
}

.email-list.invalid li {
  color: #ff4b4b;
  font-family: 'Courier New', monospace;
  background: rgba(255, 75, 75, 0.05);
  padding: 10px;
  margin-bottom: 8px;
  border-radius: 6px;
  list-style-type: none;
  border: 1px solid rgba(255, 75, 75, 0.2);
  transition: all 0.2s;
}

.email-list.invalid li:hover {
  background: rgba(255, 75, 75, 0.1);
  transform: translateX(5px);
}

.domain-list li {
  background: rgba(255, 255, 255, 0.1);
  margin: 6px;
  padding: 6px 15px;
  border-radius: 20px;
  display: inline-block;
  font-size: 0.9em;
  border: 1px solid rgba(0, 210, 255, 0.3);
  color: #a8c0ff;
  transition: all 0.2s;
}

.domain-list li:hover {
  background: rgba(0, 210, 255, 0.2);
  transform: scale(1.05);
}

.reverse-item {
  margin-bottom: 20px;
  padding-bottom: 20px;
  border-bottom: 1px dashed rgba(255, 255, 255, 0.1);
}

.reverse-item:last-child {
  border-bottom: none;
}

.code-text {
  display: block;
  background: rgba(0, 0, 0, 0.4);
  color: #00ff88;
```

```
padding: 15px;
border-radius: 6px;
font-family: 'Courier New', monospace;
word-break: break-all;
margin-top: 10px;
font-size: 0.95em;
border: 1px solid rgba(0, 255, 136, 0.1);
box-shadow: inset 0 0 10px rgba(0, 0, 0, 0.5);
}

.error {
color: #ff4b4b;
font-weight: bold;
text-align: center;
text-shadow: 0 0 5px rgba(255, 75, 75, 0.5);
background: rgba(255, 75, 75, 0.1);
padding: 10px;
border-radius: 6px;
}
```


Output

Reverse Paragraph

Enter Text to Reverse:

hello world welcome to sit,nagpur

REVERSE TEXT

REVERSED OUTPUT

Full Reverse:

rupqan,tin ot emoclew dirow olleh

Vowel Counter

Enter Paragraph to Count Vowels:

Type your paragraph here...

COUNT VOWELS

Name: M HUSSAIN KHAN
PRN: 8482871166

Email Validator & Text Reverser

Email Validation

Enter Emails (comma or space separated):

hussain.khan.batch2024@sitnagpur.siu.edu.in

VALIDATE EMAILS

ANALYSIS RESULTS

Valid: 1 Invalid: 0

Valid Emails:

hussain.khan.batch2024@sitnagpur.siu.edu.in

Domains:

sitnagpur.siu.edu.in (1)

Reverse Paragraph

Enter Text to Reverse:

Type your paragraph here...

REVERSE TEXT

Name: M HUSSAIN KHAN
PRN: 8482871166

Case Study Title

Paragraph reversal, vowel checker and email verification

Case Study Program Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Detector</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="container">
    <h2>Email Validator & Text Reverser</h2>

    <div class="tool-section">
      <h3>Email Validation</h3>
      <div class="input-group">
        <label for="emailInput">Enter Emails (comma or space separated):</label>
        <input type="text" id="emailInput" placeholder="user@example.com,
test@domain.org">
      </div>
      <button onclick="processEmails()" class="btn-email">Validate Emails</button>
      <div id="emailResults" class="result-box"></div>
    </div>

    <hr class="divider">

    <div class="tool-section">
      <h3>Reverse Paragraph</h3>
      <div class="input-group">
        <label for="reverselInput">Enter Text to Reverse:</label>
        <input type="text" id="reverselInput" placeholder="Type your paragraph
here...">
      </div>
      <button onclick="reverseText()" class="btn-reverse">Reverse Text</button>
      <div id="reverseResults" class="result-box"></div>
    </div>

    <hr class="divider">

    <div class="tool-section">
      <h3>Vowel Counter</h3>
      <div class="input-group">
        <label for="vowelInput">Enter Paragraph to Count Vowels:</label>
        <input type="text" id="vowelInput" placeholder="Type your paragraph
here...">
      </div>
      <button onclick="countVowels()" class="btn-email" style="background: linear-
gradient(45deg, #11998e, #38ef7d);">Count Vowels</button>
      <div id="vowelResults" class="result-box" style="border-left-color: #38ef7d;
background: linear-gradient(90deg, rgba(56, 239, 125, 0.1), transparent);"></div>
    </div>
  </div>
```

```

<script>
  const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;

  function processEmails() {
    const input = document.getElementById('emailInput').value;
    const resultsDiv = document.getElementById('emailResults');

    resultsDiv.innerHTML = "";
    resultsDiv.style.display = 'none';

    if (!input.trim()) {
      resultsDiv.innerHTML = '<p class="error">Please enter email addresses.</p>';
      resultsDiv.style.display = 'block';
      return;
    }

    const validEmails = [];
    const invalidEmails = [];
    const potentialEmails = input.split(/\s,+/);

    potentialEmails.forEach(email => {
      if (email.trim() === "") return;
      if (emailRegex.test(email)) {
        validEmails.push(email);
      } else {
        invalidEmails.push(email);
      }
    });

    const domainCounts = {};
    validEmails.forEach(email => {
      const domain = email.split('@')[1];
      domainCounts[domain] = (domainCounts[domain] || 0) + 1;
    });

    let output = '<h4>Analysis Results</h4>';
    output += '<div class="stat-grid">';
    output += '<div class="stat-item"><strong>Valid:</strong> <span class="count-green">' + validEmails.length + '</span></div>';
    output += '<div class="stat-item"><strong>Invalid:</strong> <span class="count-red">' + invalidEmails.length + '</span></div>';
    output += '</div>';

    if (validEmails.length > 0) {
      output += '<p><strong>Valid Emails:</strong></p><ul class="email-list valid">';
      validEmails.forEach(e => output += '<li>' + e + '</li>');
      output += '</ul>';
    }

    if (invalidEmails.length > 0) {
      output += '<p><strong>Invalid Entries:</strong></p><ul class="email-list invalid">';
      invalidEmails.forEach(e => output += '<li>' + e + '</li>');
      output += '</ul>';
    }

    if (Object.keys(domainCounts).length > 0) {
      output += '<p><strong>Domains:</strong></p><ul class="domain-list">';
      for (const domain in domainCounts) {
        output += '<li>' + domain + ' (' + domainCounts[domain] + ')</li>';
      }
      output += '</ul>';
    }

    resultsDiv.innerHTML = output;
  }

```

```

    resultsDiv.style.display = 'block';
}

function reverseText() {
    const input = document.getElementById('reverseInput').value;
    const resultsDiv = document.getElementById('reverseResults');

    resultsDiv.innerHTML = "";
    resultsDiv.style.display = 'none';

    if (!input.trim()) {
        resultsDiv.innerHTML = '<p class="error">Please enter text to reverse.</p>';
        resultsDiv.style.display = 'block';
        return;
    }

    const reversedString = input.split("").reverse().join("");

    let output = '<h4>Reversed Output</h4>';
    output += '<div class="reverse-item"><strong>Full Reverse:</strong><br><span
class="code-text">' + reversedString + '</span></div>';

    resultsDiv.innerHTML = output;
    resultsDiv.style.display = 'block';
}

function countVowels() {
    const input = document.getElementById('vowelInput').value;
    const resultsDiv = document.getElementById('vowelResults');

    resultsDiv.innerHTML = "";
    resultsDiv.style.display = 'none';

    if (!input.trim()) {
        resultsDiv.innerHTML = '<p class="error">Please enter text to count
vowels.</p>';
        resultsDiv.style.display = 'block';
        return;
    }

    const vowels = input.match(/[aeiou]/gi);
    const vowelCount = vowels ? vowels.length : 0;

    const vowelCounts = { a: 0, e: 0, i: 0, o: 0, u: 0 };
    input.toLowerCase().split("").forEach(char => {
        if (vowelCounts.hasOwnProperty(char)) {
            vowelCounts[char]++;
        }
    });

    let output = '<h4>Vowel Analysis</h4>';
    output += '<div class="stat-grid">';
    output += '<div class="stat-item"><strong>Total Vowels:</strong> <span
class="count-green">' + vowelCount + '</span></div>';
    output += '</div>';

    output += '<p><strong>Breakdown:</strong></p><ul class="domain-list">';
    for (const vowel in vowelCounts) {
        output += '<li>' + vowel.toUpperCase() + ': ' + vowelCounts[vowel] + '</li>';
    }
    output += '</ul>';

    resultsDiv.innerHTML = output;
    resultsDiv.style.display = 'block';
}
</script>

```

```
</body>
</html>
```

```
body {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  background: linear-gradient(135deg, #0f0c29, #302b63, #24243e);
  margin: 0;
  padding: 40px 20px;
  color: #e0e0e0;
  min-height: 100vh;
}

.container {
  max-width: 800px;
  margin: 0 auto;
  background: rgba(255, 255, 255, 0.05);
  backdrop-filter: blur(10px);
  padding: 40px;
  border-radius: 20px;
  box-shadow: 0 8px 32px 0 rgba(0, 0, 0, 0.37);
  border: 1px solid rgba(255, 255, 255, 0.1);
}

h2 {
  text-align: center;
  color: #ffffff;
  text-shadow: 0 0 10px rgba(255, 255, 255, 0.5);
  margin-top: 0;
  letter-spacing: 2px;
}

h3 {
  color: #a8c0ff;
  margin-bottom: 15px;
  font-size: 1.4em;
  text-shadow: 0 0 5px rgba(168, 192, 255, 0.3);
}

.tool-section {
  margin-bottom: 20px;
}

.input-group {
  margin-bottom: 20px;
}

label {
  font-weight: 600;
  display: block;
  margin-bottom: 10px;
  color: #bdc3c7;
  letter-spacing: 1px;
}

input[type="text"] {
  width: 100%;
  padding: 15px;
  border: 1px solid rgba(255, 255, 255, 0.2);
  border-radius: 8px;
  font-size: 16px;
  box-sizing: border-box;
  transition: all 0.3s;
  background: rgba(0, 0, 0, 0.3);
  color: #fff;
}
```

```

input[type="text"]:focus {
    border-color: #00d2ff;
    outline: none;
    box-shadow: 0 0 15px rgba(0, 210, 255, 0.3);
    background: rgba(0, 0, 0, 0.5);
}

button {
    width: 100%;
    padding: 15px;
    border: none;
    border-radius: 8px;
    font-size: 16px;
    font-weight: bold;
    cursor: pointer;
    transition: all 0.3s, transform 0.1s;
    color: white;
    text-transform: uppercase;
    letter-spacing: 1.5px;
    position: relative;
    overflow: hidden;
}

button:active {
    transform: scale(0.98);
}

.btn-email {
    background: linear-gradient(45deg, #00c6ff, #0072ff);
    box-shadow: 0 4px 15px rgba(0, 114, 255, 0.4);
}

.btn-email:hover {
    background: linear-gradient(45deg, #0072ff, #00c6ff);
    box-shadow: 0 6px 20px rgba(0, 114, 255, 0.6);
    transform: translateY(-2px);
}

.btn-reverse {
    background: linear-gradient(45deg, #f12711, #f5af19);
    box-shadow: 0 4px 15px rgba(241, 39, 17, 0.4);
}

.btn-reverse:hover {
    background: linear-gradient(45deg, #f5af19, #f12711);
    box-shadow: 0 6px 20px rgba(241, 39, 17, 0.6);
    transform: translateY(-2px);
}

.divider {
    border: 0;
    height: 1px;
    background: linear-gradient(90deg, transparent, rgba(255, 255, 255, 0.3),
transparent);
    margin: 50px 0;
}

.result-box {
    margin-top: 30px;
    padding: 25px;
    border-radius: 12px;
    background: rgba(0, 0, 0, 0.2);
    border-left: 5px solid #ccc;
    display: none;
    animation: fadeIn 0.5s ease;
}

```

```

    backdrop-filter: blur(5px);
}

#emailResults {
    border-left-color: #00d2ff;
    background: linear-gradient(90deg, rgba(0, 210, 255, 0.1), transparent);
    box-shadow: 0 0 15px rgba(0, 210, 255, 0.1);
}

#reverseResults {
    border-left-color: #f5af19;
    background: linear-gradient(90deg, rgba(245, 175, 25, 0.1), transparent);
    box-shadow: 0 0 15px rgba(245, 175, 25, 0.1);
}

@keyframes fadeIn {
    from { opacity: 0; transform: translateY(10px); }
    to { opacity: 1; transform: translateY(0); }
}

h4 {
    margin-top: 0;
    color: #fff;
    border-bottom: 1px solid rgba(255, 255, 255, 0.1);
    padding-bottom: 15px;
    text-transform: uppercase;
    font-size: 0.9em;
    letter-spacing: 1px;
}

.stat-grid {
    display: flex;
    justify-content: space-around;
    margin-bottom: 20px;
    background: rgba(255, 255, 255, 0.05);
    padding: 15px;
    border-radius: 8px;
    box-shadow: 0 4px 15px rgba(0, 0, 0, 0.2);
    border: 1px solid rgba(255, 255, 255, 0.05);
}

.stat-item {
    text-align: center;
    font-size: 1.1em;
    color: #bdc3c7;
}

.count-green {
    color: #00ff88;
    font-weight: bold;
    text-shadow: 0 0 10px rgba(0, 255, 136, 0.4);
}

.count-red {
    color: #ff4b4b;
    font-weight: bold;
    text-shadow: 0 0 10px rgba(255, 75, 75, 0.4);
}

ul {
    padding-left: 0;
    margin: 15px 0;
}

.email-list.valid li {
    color: #00ff88;

```

```

    font-family: 'Courier New', monospace;
    background: rgba(0, 255, 136, 0.05);
    padding: 10px;
    margin-bottom: 8px;
    border-radius: 6px;
    list-style-type: none;
    border: 1px solid rgba(0, 255, 136, 0.2);
    transition: all 0.2s;
}

.email-list.valid li:hover {
    background: rgba(0, 255, 136, 0.1);
    transform: translateX(5px);
}

.email-list.invalid li {
    color: #ff4b4b;
    font-family: 'Courier New', monospace;
    background: rgba(255, 75, 75, 0.05);
    padding: 10px;
    margin-bottom: 8px;
    border-radius: 6px;
    list-style-type: none;
    border: 1px solid rgba(255, 75, 75, 0.2);
    transition: all 0.2s;
}

.email-list.invalid li:hover {
    background: rgba(255, 75, 75, 0.1);
    transform: translateX(5px);
}

.domain-list li {
    background: rgba(255, 255, 255, 0.1);
    margin: 6px;
    padding: 6px 15px;
    border-radius: 20px;
    display: inline-block;
    font-size: 0.9em;
    border: 1px solid rgba(0, 210, 255, 0.3);
    color: #a8c0ff;
    transition: all 0.2s;
}

.domain-list li:hover {
    background: rgba(0, 210, 255, 0.2);
    transform: scale(1.05);
}

.reverse-item {
    margin-bottom: 20px;
    padding-bottom: 20px;
    border-bottom: 1px dashed rgba(255, 255, 255, 0.1);
}

.reverse-item:last-child {
    border-bottom: none;
}

.code-text {
    display: block;
    background: rgba(0, 0, 0, 0.4);
    color: #00ff88;
    padding: 15px;
    border-radius: 6px;
    font-family: 'Courier New', monospace;

```



```
word-break: break-all;
margin-top: 10px;
font-size: 0.95em;
border: 1px solid rgba(0, 255, 136, 0.1);
box-shadow: inset 0 0 10px rgba(0, 0, 0, 0.5);
}
```

```
.error {
  color: #ff4b4b;
  font-weight: bold;
  text-align: center;
  text-shadow: 0 0 5px rgba(255, 75, 75, 0.5);
  background: rgba(255, 75, 75, 0.1);
  padding: 10px;
  border-radius: 6px;
}
```

Output

The image displays two screenshots of a web application interface, likely a browser window, showing the output of various text processing functions. The interface has a dark blue background with white and light blue text and buttons.

Top Screenshot:

- Email Validator & Text Reverser**
- Email Validation**
Enter Emails (comma or space separated):
user@example.com, test@domain.org
VALIDATE EMAILS
- Reverse Paragraph**
Enter Text to Reverse:
Type your paragraph here...
REVERSE TEXT
- Vowel Counter**
Enter Paragraph to Count Vowels:
Type your paragraph here...
COUNT VOWELS

Bottom Screenshot:

- Full Reverse:**
rupgan,tis ot emoclew dirow olleh
- Vowel Counter**
Enter Paragraph to Count Vowels:
java script is cool language
COUNT VOWELS
- VOWEL ANALYSIS**
Total Vowels: 10
Breakdown:
A: 4 E: 1 I: 2 O: 2 U: 1

Both screenshots show a yellow box in the bottom right corner with the text: Name: M HUSSAIN KHAN, PRN: 8482871166.

Result/Conclusion

The program was executed successfully. JavaScript string functions and regular expressions were used to validate email addresses, extract required information, and perform text analysis effectively.